LAB - 08 DATE:-30|05|2026

Q1. Write a program to reverse a string without using built-in reverse functions.

Lab_08 > Q1.java > Q1

```java
package Lab_08;

public class Q1 {
    Run | Debug
    public static void main(String[] args) {
        String str = "Hello";
        String rev = "";

        for (int i = str.length() - 1; i >= 0; i--) {
            rev += str.charAt(i);
        }

        System.out.println("Reversed: " + rev);
    }
}
```

OUTPUT:—

```
gauravmathur@Annus-Mac LabJava % javac -d . Lab_08/Q1.java
gauravmathur@Annus-Mac LabJava % java Lab_08/Q1
Reversed: olleH
```

2. Write a program to count the number of words in a string (with and without split() method).

Lab_08 > Q2.java > Q2

```java
package Lab_08;

public class Q2 {
    Run | Debug
    public static void main(String[] args) {
        String str = "Hello world Java";
        int count = 1;

        for (int i = 0; i < str.length(); i++) {
            if (str.charAt(i) == ' ') count++;
        }

        System.out.println("Words: " + count);
    }
}
```

OUTPUT:—

```
gauravmathur@Annus-Mac LabJava % javac -d . Lab_08/Q2.java
gauravmathur@Annus-Mac LabJava % java Lab_08/Q2
Words: 3
```

Q3. Create a program **to** remove duplicate characters from a string.

Lab_08 > Q3.java > Q3

```java
package Lab_08;

public class Q3 {
    Run | Debug
    public static void main(String[] args) {
        String str = "programming";
        String result = "";
        for (int i = 0; i < str.length(); i++) {
            char ch = str.charAt(i);

            if (result.indexOf(ch) == -1) {
                result += ch;
            }
        }

        System.out.println("Result: " + result);
    }
}
```

OUTPUT:—

```
gauravmathur@Annus-Mac LabJava % javac -d . Lab_08/Q3.java
gauravmathur@Annus-Mac LabJava % java Lab_08/Q3
Result: progamin
```

Q4. Write a program **to** find the frequency of each character in a string.

Lab_08 > Q4.java > Q4

```java
package Lab_08;

public class Q4 {
    Run | Debug
    public static void main(String[] args) {
        String str = "hello";

        for (int i = 0; i < str.length(); i++) {
            char ch = str.charAt(i);
            int count = 0;

            for (int j = 0; j < str.length(); j++) {
                if (str.charAt(j) == ch) count++;
            }

            if (str.indexOf(ch) == i)
                System.out.println(ch + " : " + count);
        }
    }
}
```

OUTPUT:—

```
gauravmathur@Annus-Mac LabJava % javac -d . Lab_08/Q4.java
gauravmathur@Annus-Mac LabJava % java Lab_08/Q4
h : 1
e : 1
l : 2
o : 1
```

Q5. Create a program to capitalize the first letter of each word in a sentence.

Lab_08 > Q5.java > Q5

```java
package Lab_08;

public class Q5 {
    Run | Debug
    public static void main(String[] args) {
        String str = "hello world java";
        String[] words = str.split(regex: " ");
        String result = "";
        for (String word : words) {
            result += Character.toUpperCase(word.charAt(index: 0)) + word.substring(beginIndex: 1) + " ";
        }

        System.out.println(result);
    }
}
```

OUTPUT:—

```
gauravmathur@Annus-Mac LabJava % javac -d . Lab_08/Q5.java
gauravmathur@Annus-Mac LabJava % java Lab_08/Q5
Hello World Java
```

Q6.Write a program to create three strings and compare each pair of strings using the `compareTo()` method. Display the numeric values returned by each comparison.

Lab_08 > Q6.java > Q6

```java
package Lab_08;

public class Q6 {
    Run | Debug
    public static void main(String[] args) {
        String s1 = "apple";
        String s2 = "banana";
        String s3 = "apple";

        System.out.println(s1.compareTo(s2));
        System.out.println(s2.compareTo(s3));
        System.out.println(s1.compareTo(s3));
    }
}
```

OUTPUT:—

```
gauravmathur@Annus-Mac LabJava % javac -d . Lab_08/Q6.java
gauravmathur@Annus-Mac LabJava % java Lab_08/Q6
-1
1
0
```

Q7Write a program to implement your own version of `equals()` method for string comparison (without using equals() or compareTo()).

Lab_08 > Q7.java > Q7

```java
package Lab_08;

public class Q7 {
    Run | Debug
    public static void main(String[] args) {
        String s1 = "hello";
        String s2 = "hello";
        boolean isEqual = true;
        if (s1.length() != s2.length()) {
            isEqual = false;
        } else {
            for (int i = 0; i < s1.length(); i++) {
                if (s1.charAt(i) != s2.charAt(i)) {
                    isEqual = false;
                    break;
                }
            }
        }

        System.out.println("Equal: " + isEqual);
    }
}
```

OUTPUT:—

```
gauravmathur@Annus-Mac LabJava % javac -d . Lab_08/Q7.java
gauravmathur@Annus-Mac LabJava % java Lab_08/Q7
Equal: true
```

Q8. Write a program to compare one string created using a **literal** and another using the **`new`** keyword using both == and `equals()`.

Lab_08 > Q8.java > Q8

```java
package Lab_08;

public class Q8{
    Run | Debug
    public static void main(String[] args) {
        String s1 = "hello";
        String s2 = new String(original: "hello");

        System.out.println("Using == : " + (s1 == s2));          // false
        System.out.println("Using equals(): " + s1.equals(s2)); // true
    }
}
```

OUTPUT:—

```
Equal: true
gauravmathur@Annus-Mac LabJava % javac -d . Lab_08/Q8.java
gauravmathur@Annus-Mac LabJava % java Lab_08/Q8
Using == : false
Using equals(): true
```